



**MINNESOTA STATE UNIVERSITY, MANKATO
DEPARTMENT OF COMPUTER INFORMATION SCIENCE
COMPUTER SCIENCE PROGRAM**

REQUIREMENTS ANALYSIS DOCUMENT

THE MavPASS SCHEDULING PROJECT

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TABLE OF CONTENTS

Title Page	1
Table of Contents	2
Introduction	3
Purpose	3
Overview	3
Scope	3
Current System	4
Overview	4
Scheduling Constraints	4
Proposed System	5
Overview	5
Functional Requirements	5
Non-functional Requirements	6
Definitions, Acronyms, and Abbreviations	6
References	6

1. INTRODUCTION

1.1 PURPOSE

This document outlines the requirements for the MavPASS Scheduling product that need to be met for the project to be considered successful. It also documents the findings after careful analyses of the clients' current system were undertaken. The intended audience of this document is anyone who is involved in the project team.

1.5 OVERVIEW

This project aims to develop and provide MavPASS with an efficient, convenient, easy-to-use scheduling solution to replace their current mode of scheduling to allow them to expand and support more students across courses at Minnesota State University, Mankato.

1.2 SCOPE

Deliverables	Type of work	Activities	Resources	Tech Skills	Priority
Requirements analysis document: meet with Laura and Lina (and possibly student MavPASS leaders) to understand clearly what they think a "winning" solution will be. Study and absorb the extensive documentation they have prepared that describes their current scheduling process. Document, in a simple and concise fashion, all the application requirements including data input, option/preference specification, and results output.	Requirements analysis and study of existing process. Documentation – in simple/concise format – of requirements for automated or semi-automated solution.	Requirements gathering meetings, study of existing scheduling process, drafting requirements document, getting feedback from clients and incorporating changes they request until agreement is reached.	Access to client leads, documentation existing process	Spreadsheets, word processing	High
User stories that capture and span intended use cases and requirements.	Writing user stories	Writing user stories and presenting them to client for feedback, making modifications as needed	Document creation tool to be agreed with client	Familiarity with user story approach and techniques	High
Acquire and clean full example sets of data to be used for development and testing. This is likely to include multiple sources.	Data analysis and data cleansing	Work with client to obtain and get approval on data sets to be used for development and testing	Access to client lead, data provided by client, analysis assistance from client	Data management	High
Wireframe mockup of user interface and user experience. Be sure to include how users will set up data input, specification of preferences and options, and, of course, delivery of resulting schedules.	UI mockup	Design and create wireframe version of user interface and full-application user experience, manage iterative feedback from client to achieve desired solution	Wireframe tool of team's choice	Use of selected wireframe tool	High
Design document for application. Special attention to be paid to designing the application so that it can be modified, updated, and maintained by client after the project is complete. (In other words, a full software lifecycle model must be included in the design document.)	Software design	Detailed design for data representation, scheduling algorithm, and user interface.	Online documentation for required development tools, as selected by project team and approved by client	Programming language and front end development tools to be chosen by team and approved by client. (Pay special attention to full lifecycle maintenance here.)	High
Test document for application.	Test planning, execution, and documentation	Creating test plan, executing tests iteratively, documenting results and managing incremental improvements.	Examples of test documents from other projects, support from IT team, word processing tools	Test planning, execution, and documentation	Medium
Working application.	Coding	Implement Front End, Server layer, and Database to create full stack application.	See above	Programming language and front end development tools as previously agreed.	Medium
Detailed user documentation that includes all steps needed to make the application work as each new semester arrives.	Documentation	Documenting all steps the client user must take to adapt the application as each new semester arrives.	Close interaction with client leads, acceptance of client leads	Word processing	High

2. CURRENT SYSTEM

2.1 OVERVIEW

The current mode of scheduling for MavPASS Leaders, Rooms, and Observers occurs in 3 phases as described below.

Phase 1: Collecting MavPASS Leader's schedules and putting them in the spreadsheet and collecting room schedules.

Phase 2: Creating the second spreadsheet with the colored bar graph of student's availability

Phase 3: Using all spreadsheets as well as room schedules to make the master schedule for MavPASS Sessions.

This process is extremely time-consuming and inefficient. It also adds lots of opportunity for human error which can cause sessions to be disrupted or needing to change rooms last second.

2.2 SCHEDULING CONSTRAINTS

Soft Constraints

- MavPASS Leaders cannot have two sessions on the same day
- MavPASS sessions should be close to the original class's room and time

Hard Constraints

A. MavPASS Leaders

- Cannot have two sessions at the same time
- Can only have between 2-3 regular sessions per week
- Sessions cannot be scheduled when the leader is unavailable
- Two leaders can't have a session at the same time in the same room

B. MavPASS Rooms

- Cannot be scheduled when it is unavailable
- Cannot have two sessions at the same time
- Must meet the capacity requirements of the leader's course
- Must have computers if being used for CS
- Must have a whiteboard
- Anatomy needs to have access to the bone room

3. PROPOSED SYSTEM

3.1 OVERVIEW

Our proposed system would be a web-based interface for MavPASS Coordinators to better manage scheduling by allowing MavPASS Leaders to enter their availability and allow the coordinators to enter the room schedule. The product would then give the MavPASS Coordinators an editable master schedule.

3.2 FUNCTIONAL REQUIREMENTS

Data Collection: MavPASS Leaders will input their availability for the semester and the number of sessions per week. MavPASS Coordinators will input the room availability for the semester. Observers will put in their availability.

Schedule Generation: A scheduling algorithm script will then go through and schedule the leader's sessions using their availability, number of sessions per week, and room availability. The master schedule will then be generated and displayed.

Observation Schedules: Observation Schedules will be handled separately from the MavPASS Leader's session schedules. The following is required out of the Observation Schedule:

- Observers can pick a course, leader, or room of which they want to observe a session
- After that initial selection the observer will be given sessions that they are available to attend
- Observers will select the sessions to observe for the week or however far in advance is specified by the coordinator's
- An Observation Schedule will be updated automatically after an observer submits their selection

Data Filtering: It is very important for the data to be filtered and viewed in the following ways.

- **Room:** A schedule can be viewed for each individual room and a button can be clicked to print/save as pdf.
- **MavPASS Leader :** A schedule can be viewed for each individual MavPASS Leader, and a button can be clicked to print/save as pdf.
- **Course:** A schedule can be viewed for each individual course and a button can be clicked to print/save as pdf.
- **Daily:** A schedule can be viewed for each day which will show all sessions between the times of 9am – 9pm which can also be printed/saved as a pdf via a button.
- **Observation:** A daily or weekly observation schedule can be viewed and printed/saved as pdf

Schedule Modification:

- **MavPASS Leader Sessions:** Coordinators should be able to easily modify the master schedule by selecting a MavPASS Leader, the session time/room that needs to change, and then the scheduling algorithm should be able to suggest available times based on room availability and the leader's availability.
- **Observations:** Observers should be able to modify when they are scheduled to observe if needed.
- **Updates:** All schedules should be updated when a coordinator makes a change.

Security:

- **Coordinators:** Should be able to modify and view all schedules.
- **Coaches:** Should be able to view all schedules but can only modify their own observation times.
- **MavPASS Leaders:** Should be able to view all leader-related schedules but cannot edit.

3.3 NON-FUNCTIONAL REQUIREMENTS

Usability: The product should be easy to learn/use by the MavPASS staff. The UI should be laid out in a very understandable way in order to draw attention to the parts that matter and should look clean and polished. The client should be satisfied with the product.

Reliability: The product should be free of bugs and the database should be hosted appropriately so that there are no third-party issues that can interfere with the product. There should be no critical failures that arise upon use. Restrictions should not be inappropriately placed on users which could cause problems with use/security.

Performance: There should be little to no wait time (no longer than a few seconds) when creating the master schedule itself using the algorithms, nor when entering/modifying data. The number of users at any given time should not be a concern for performance slowdowns.

Supportability: Through careful and well-thought-out documentation the supportability of the product should be as simple as possible since knowledge transfer would be streamlined and there would be no functionality confusion.

Implementation: The product will be a web application that ideally can be viewed on both desktop/laptop and mobile.

Interface: Accessible on the web and able to use in Google, Edge, and Firefox.

Packaging: MavPASS coordinators should have access to the full functionality of the system.

Legal: N/A

4. DEFINITIONS, ACRONYMS, AND ABBREVIATIONS

MavPASS: Maverick Peer-facilitated Academic Support System

MavPASS Coordinator: Personnel in charge

MavPASS Leader: Peer Facilitator who holds sessions and office hours weekly to assist students in the specified course using facilitation skills in peer-to-peer supplemental instructions.

5. REFERENCES

Phase 1: [Available Times for MP Sessions F22.xlsx](#)

Phase 2: [F22 MavPASS Leaders Availability Color Bar Graph F22.xlsx](#)

Phase 3: [ML45 Sessions F22.docx](#)